Simplex Algorithm for LP

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Overview

1. Formulation so far
2. The Curse of Dimensionality
3. Overview of the Simplex Algorithm
4. Choosing the leaving and entering indices

Frame Title

So far we have rewritten the LP as a search among finitely many basic feasible points

$$min_{x\in\mathfrak{B}} c^T x$$
$$x \geq 0 \ , \ \ Ax = b$$

where $\mathfrak{B}$ is the set of all basic feasible points which is finite

Curse of Dimensionality

- One naive algorithm to solve an LP would be to enumerate and search over all vertices as follows: Choose a subset $\mathcal{B}$ of $\{1, 2, \ldots, n\}$ with $|\mathcal{B} = m\}$ and solve for the equations $x_i = 0$ for $i \notin \mathcal{B}$ and $Ax = b$. Verify if x is feasible - i.e. $x \geq 0$. Then include x in the set of all basic feasible points $\mathfrak{B}$.
- After trying out all possible subsets of $\{1, 2, \ldots, n\}$ with m elements and collecting all basic feasible points, see on which of them, $c^T x$ is minimum
- But the number of subsets of $\{1, 2, \ldots, n\}$ with m elements increases exponentially with n and hence the computational time blows up soon quickly enough as n increases - this is the curse of dimensionality

Overview of the Simplex Algorithm

The simplex algorithm roughly works as follows:

- Start with an initial basic feasible point x_0
- While $i = 0, 1, \ldots,$
  Get a basic feasible point x_i , return another basic feasible x_{i+1} such that $c^T x_{i+1} < c^T x_i$
  (The objective function decreases among successive iterates of the basic feasible points)
- Stop when one cannot proceed further

The algorithm will terminate if the objective function is bounded over the feasible region and hence has a basic feasible point minimizing $c^T x$

Details of the simplex algorithm

- Given a basic feasible point x, let $\mathcal{B}$ be the index corresponding to it.
- Then, the simplex algorithm replaces exactly one index in $\mathcal{B}$ by another index not in $\mathcal{B}$ so that the basic feasible point x calculated by the replaced new $\mathcal{B}$ with the replacement, decreases the objective function
- So, the only two things to decide in the simplex algorithm are to decide
  1. which index to remove from the basis $\mathcal{B}$ - leaving index p
  2. which index not in $\mathcal{B}$ to replace it with - entering index q

KKT Conditions Recalled

$$
\begin{aligned}
\textit{Gradient Condition:} \quad & A^T\lambda + s = c \\
\textit{Feasibility:} \quad & Ax = b \\
\textit{Feasibility:} \quad & x \geq 0 \\
\textit{Inequality:} \quad & s \geq 0 \\
\textit{Complementary Slackness:} \quad & x_i s_i = 0
\end{aligned}
$$

$s_B = 0, x_N = 0$

- Let $\mathcal{N} = \{1, 2, 3, \dots, n\}/\mathcal{B}$ at a basic feasible point x in the current iteration.
- **Partitioning the variables:** Let

$$
\begin{aligned}
x &= x_B + x_N \\
s &= s_B + s_N \\
c &= c_B + c_N \\
A &= [B : N]
\end{aligned}
$$

- Since x is a basic feasible point, $x_i = 0\ \forall i \in N$ and $x \geq 0$, which yields $x_N = 0$. By KKT condition of complementary slackness, $s_B = 0$ and hence, $b = Ax = Bx_B + Nx_N = Bx$ and hence $x_B = B^{-1}b \geq 0$

Finding λ, s_N

- By the gradient condition in KKT, we have

$$\begin{aligned} B^T\lambda &= c_B \\ N^T\lambda + s_N &= c_N \end{aligned}$$

- Since B is square and non-singular, the first equation above uniquely solves for λ and hence putting it in the second condition gives [1] s_N as

$$s_N = c_N - (B^{-1}N)^T c_B$$

- If you look back, you would see that all the KKT conditions have been enforced except for $s \geq 0$. But already $s_B = 0$ and hence $s_N \geq 0$ is remaining to be satisfied.

[1] This step is called pricing

Choosing the leaving index q

- If s_N calculated above satisfied $s_N \geq 0$, then all the KKT conditions would be satisfied and hence the given feasible point is indeed the minimum (**termination condition**)
- If not, then atleast one of the components of s_N is negative. Pick the entering index q to be one of those indices - i.e.

$$q = pick - among\{q' \in \mathcal{N} \text{ such that } s'_q < 0\}$$

Choosing the entering index p - the overall idea

- Keep all other components of x_N fixed at 0 except x_q
- Since $x_q = 0$, now increase x_q to positive values until one of the components of the corresponding feasible x_B is driven to zero - call that component driven to zero as x_p (**If no component of feasible x_B is driven to zero for whatever increase in x_q, then the problem is unbounded - terminate**)
- This p is the leaving index - remove index p in $\mathcal{B}$ and replace by q

Choosing p and pivoting - precisely

- Let x^+ be the next iterate of x. x^+ differs from x only in the p and q components. Hence,

$$b = Ax = Ax^+ = Bx_B^+ + A_q x_q^+ = Bx_B$$

  since $x_q^+ = 0$ (q will be a part of B and hence not a part of N for x^+).
- So,

$$x_B^+ = x_B - B^{-1} A_q x_q^+$$

- One must choose x^+ so that one of the constraints in B must be active - i.e. $x_p = 0$ for some $p \in \mathcal{B}$.

Analyses

$$
\begin{aligned}
c^T x^+ &= c_B^T x_B^+ + c_q x_q^+ = c_B^T (x_B - B^{-1} A_q x_q^+) + c_q x_q \\
&= c_B^T x_B - c_B^T B^{-1} A_q x_q^+ + c_q x_q^+
\end{aligned}
$$

But from the blue equations, $c_B^T = \lambda^T B$ and the KKT gradient condition $A^T \lambda + s = c$, we have

$$
c^T x^+ = c_B^T x_B - (c_q - s_q) x_q^+ + c_q x_q^+ = c^T x + s_q x_q^+
$$

Since q was chosen to have $s_q < 0$, choose $x_q^+ > 0$ such that $x_p = 0$ where $x_B^+ = x_B - B^{-1} A_q x_q^+ > 0$.

As seen already, if $x_B \geq 0$ for all $x_q > 0$, then the problem is unbounded

Simplex algorithm - Summary

Procedure (One Step of Simplex).

Given $\mathcal{B}, \mathcal{N}, x_B = B^{-1}b \geq 0, x_N = 0$;
Solve $B^T\lambda = c_B$ for λ,
Compute $s_N = c_N - N^T\lambda$; (* pricing *)
if $s_N \geq 0$
  stop; (* optimal point found *)
Select $q \in \mathcal{N}$ with $s_q < 0$ as the entering index;
Solve $Bd = A_q$ for d;
if $d \leq 0$
  stop; (* problem is unbounded *)
Calculate $x_q^+ = \min_{i \mid d_i > 0} (x_B)_i / d_i$, and use p to denote the minimizing i;
Update $x_B^+ = x_B - dx_q^+, x_N^+ = (0, \ldots, 0, x_q^+, 0, \ldots, 0)^T$;
Change $\mathcal{B}$ by adding q and removing the basic variable corresponding to column p of B.

Figure: Simplex Algorithm for a single iteration

Worked Example

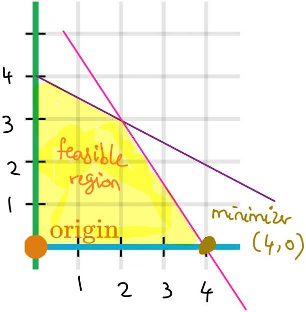
$$\min_{x \in \mathbb{R}^4} \quad -x_1 - x_2$$

Sub $\quad x_1 \geq 0, \; x_2 \geq 0$

$$x_1 + 2x_2 \leq 8$$

$$3x_1 + 2x_2 \leq 12$$

<u>Geometry</u>



Worked Example

Rewriting as standard problem

$$\min_{x_1, x_2, x_3, x_4} \quad \underbrace{-x_1 - x_2}_{c = [-1 \ -1 \ 0 \ 0]}$$

Sub

$$x_1 + 2x_2 + x_3 = 8$$
$$3x_1 + 2x_2 + x_4 = 12$$

$$x_1, x_2, x_3, x_4 \geq 0$$

$$\underbrace{\begin{bmatrix} 1 & 2 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 8 \\ 12 \end{bmatrix}}_{b}$$

full rank!

Worked Example

Initial : $B = \{3,4\}$ $N = \{1,2\}$

$x_B = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}$ $\quad x_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$s_B = \begin{bmatrix} s_3 \\ s_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad s_N = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}$

$c_B = \begin{bmatrix} c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$

$c_N = \begin{bmatrix} c_1 & c_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \end{bmatrix}$

$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = B^{-1}$ $\quad N = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

$B^T \lambda = c_B = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \lambda = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Worked Example

$$x_B = B^{-1} b = I b = \begin{bmatrix} 8 \\ 12 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}$$

<u>Computation of S_N</u>

$$S_N = C_N - (B^{-1}N)^T C_B$$

$$= \begin{bmatrix} -1 \\ -1 \end{bmatrix} - (B^{-1}N)^T C_B \rightarrow = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

S_1, S_2 both < 0

Hence entering index $= 1$ or 2

choose $q = 1$, $A_q = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

Worked Example

Computation of x_B^+

$$c^T x^+ = c^T x + \underbrace{\tilde{s}_q}_{s_1 = -1} \underbrace{x_q^+}_{=1}$$

$$c^T x = \begin{bmatrix} -1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 8 \\ 12 \end{bmatrix} = 0$$

(originally)

$$\Rightarrow \boxed{c^T x^+ = -x_1^+}$$

$$c^T x - s_q x_q^+ = -s_q x_q^+$$

$$x_B^+ = x_B - \left(B^{-1} A_q x_q^+\right)$$

$$= \begin{bmatrix} 8 \\ 12 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \end{bmatrix} x_q^+$$

$$= \begin{bmatrix} 8 - x_q^+ \\ 12 - 3x_q^+ \end{bmatrix}$$

Worked Example

Choose $x_q^+ = 4 \Rightarrow$

$$x_B^+ = \begin{bmatrix} 8-4 \\ 12-(3\times 4) \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$x_4 = 0$

So, in next iterate

$$B = \{3, 1\} = \{1, 3\}$$

replaced

$$N = \{4, 2\} = \{2, 4\}$$

$$c_N = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 1 \\ 3 & 0 \end{bmatrix}$$

$$N = \begin{bmatrix} 2 & 0 \\ 2 & 1 \end{bmatrix} \qquad c_B = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Worked Example

$$s_N = c_N - (B^{-1}N)^T c_B$$

$$= \begin{bmatrix} -0.333 \\ 0.333 \end{bmatrix} \leftarrow s_2$$

So, entering index $q = 2$

$$A_q = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$x_B = B^{-1}b = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

$$x_B^+ = x_B - B^{-1}A_q x_q^+$$

$$\begin{bmatrix} x_4 \\ x_3 \end{bmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} - B^{-1}A_q x_q^+$$

$$= \begin{pmatrix} 4 - 0.667\, x_q^+ \\ 4 - 1.333\, x_q^+ \end{pmatrix} \begin{matrix} = 0 \Rightarrow x_q^+ = 6 \\ = 0 \Rightarrow x_q^+ = 3 \end{matrix}$$

Worked Example

choosing lower value $x_q^+ = 3$

$\Rightarrow x_3 = 3$

$\Rightarrow x_3 = 0 \Rightarrow$ leaving index $= 3$

new:

So, $B = \{1, 3\} \longrightarrow B = \{1, 2\}$

$N = \{2, 4\} \longrightarrow N = \{3, 4\}$

<u>Next iteration</u>

$B = \{1, 2\}$ $\qquad c_B = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

$N = \{3, 4\}$ $\qquad c_N = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$B = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \qquad N = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$S_N = c_N - (B^{-1}N)^T c_B$

Worked Example

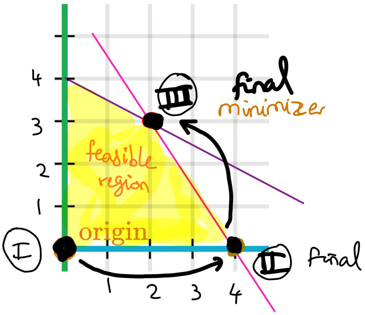
$$S_N = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - (B^{-1} I)^T \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{matrix} \rightarrow \geq 0 \\ \rightarrow \geq 0 \end{matrix}$$

iteration stops

Worked Example

Geometrically,



$c^T x$: I : 0

II : -4

III : -5